Biomedical Nanotechnology
(Assignment-1)

Answer ALL Questions (10 x 1 = 10 marks)

Q1. Choose the appropriate characteristics of incidental nanoparticles

(a) different shape and uniform size
(b) uniform shape and size
(c) **composed of different elements and non-uniform size**
(d) controlled shape and size with organic coating

Q2. Assertion(A)- Energy savings windows designed to stop thermal radiation escaping in winter and prevent the entry of infrared radiation from the sun entering the building

Reason(R)- The window glass is coated with thin film (5-10 nm) of vanadium dioxide

(a) **Both A and R are true, and R is the correct explanation of A.**
(b) Both A and R are true, but R is NOT the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true

Q3. Cowpea chlorotic mottle virus (CCMV)

(i) possesses chemically distinct environment (internally and externally)
(ii) interior surface is less positively charged than the exterior
(iii) interior surface is more positively charged than the exterior
(iv) exterior surface serves as a nucleation site for crystal growth

Choose the correct option

(a) (i) and (iii)
(b) (ii) and (iv)
(c) (i), (ii) and (iv)
(d) (ii), (iii) and (iv)

Q4. Which of the following is not related to "stern layer"

(a) Known as electrical double layer
(b) Rigid layer of ions tightly bound to the particle
(c) Ions bound to the stern layer travel with the particles
(d) slipping plane is the boundary of the stern layer

Q5. Nanomaterials properties differ from their same mass of material because

(a) Conduction band and unoccupied band lies close to each other
(b) Conduction band and unoccupied band separated from each other
(c) Electrons motion becomes confined
(d) **Both (b) and (c)**

Q6. Densely packed nanoscale pillars structures kill bacteria by

(a) Releasing antibiotics
(b) Illuminating UV-rays
(c) **Piercing the membrane of bacterial cells on contact and tearing bacteria apart**
(d) Both (a) and (b)

Q7. The role of citrate and thiols in metal nanoparticles synthesis is to

(a) Prevent the reduction of metal salts
(b) accelerate the reduction process of metal salts
(c) **prevent the agglomeration of metal nanoparticles**
(d) carry out the oxidation of metal salts

Q8. Materials and setup required for silver nanoparticles synthesis by laser ablation methods

(a) Silver nitrate, Nd YAG laser, mirror, concave lens, ultra-pure water, CTAB
(b) Silver nitrate, Nd YAG laser, mirror, ultra-pure water, x-y-z stage, $LiAlH_4$
(c) **Silver slice, Nd YAG laser, mirror, convex lens, ultra-pure water, x-y-z stage**
(d) Silver slice, Nd YAG laser, mirror, concave lens, ultra-pure water, x-y-z stage

Q9. Biosynthesis of Cadmium sulphide nanocrystals done by

(a) *Candida glabrata*
(b) *Schizosaccharomyces pombe*
(c) *Verticillium*
(d) **Both *Candida glabrata* and *Schizosaccharomyces pombe***

Q10. Assertion(A): The AFM measures the force of attraction between a probe and sample surface.

Reason(R): Works only on the conducting samples

(a) Both Assertion and reason are correct
(b) Only Assertion is correct
(c) Both assertion and reason are incorrect
(d) Only reason is correct